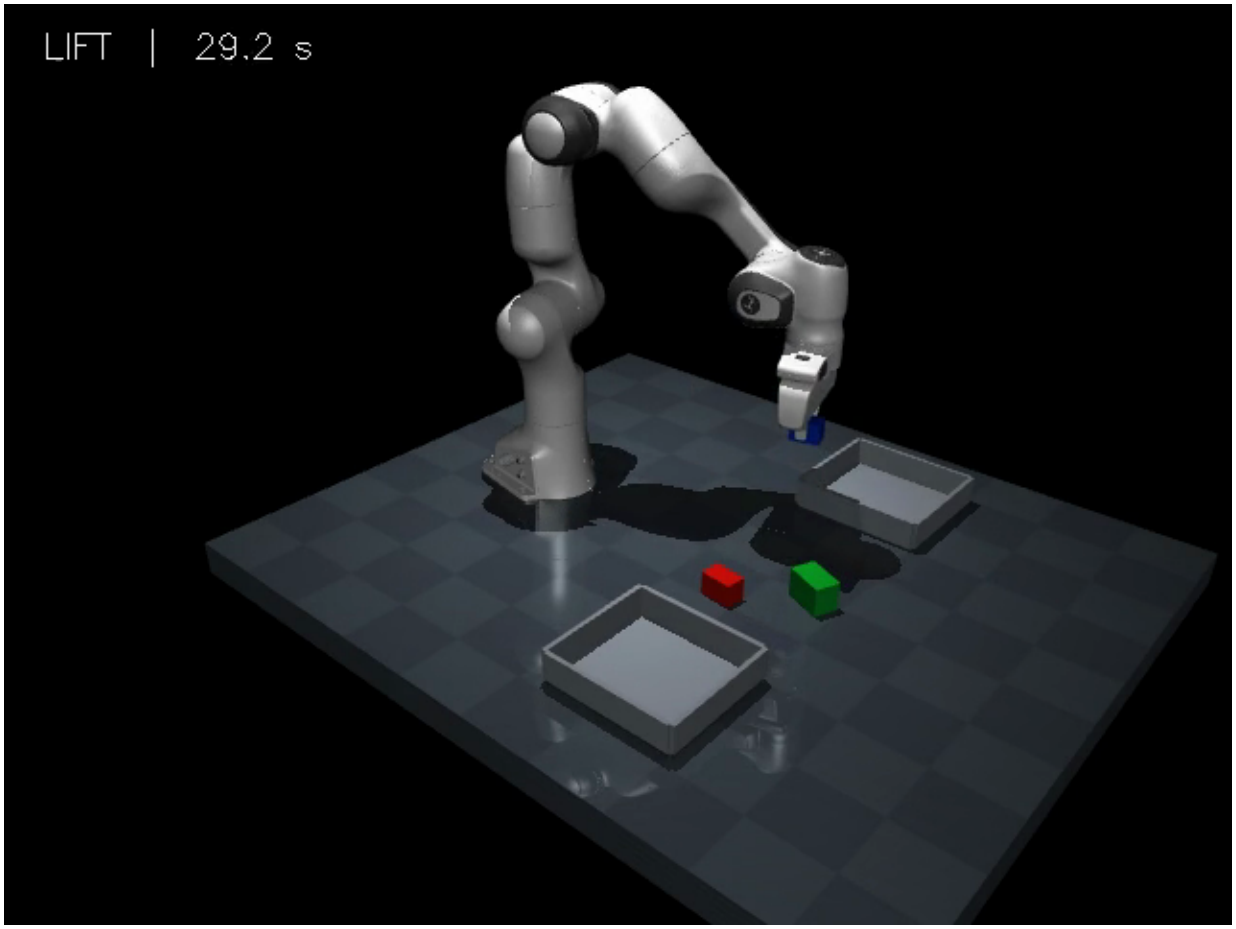
Camera-Guided Robot Arm Sorting

FINAL SIMULATION PORTFOLIO / 10 OCTOBER 2026

An operator says “Pick up the blue component,” then “Left.” The workcell connects that request to camera pixels, grasps through physical contact, and checks placement before reporting completion.

Frozen integrated protocol	Completed placement cases	Unexpected collisions
15/15 declared checks	10/10	0



Exact sampled overview frame near the end of the final reserved blue pickup. MuJoCo simulation; no physical robot. Source frame and hash are retained with the final film.

The result is a reproducible known-part workcell, with an honest development story and measured failure behavior. Pretrained image-language selection, engineered supervision and learned robot kinematics have distinct responsibilities.

How the workcell makes a decision

Stage	Responsibility and boundary
Request and image	Qwen3-VL-2B receives actual numbered RGB pixels and the object reference. No simulator object inventory or true poses are supplied.
Validate and localize	Schema, destination, colour, geometry and image-space rank checks authorize one known target. RGB-D depth/calibration estimate its pose. Duplicate colours require clarification.
Execute	Learned 6-by-7 Jacobian supports conventional feedback. Goals, gripper sequencing, joint/rate limits, phase checks and bounded retry are engineered.
Observe and supervise	Separate ROS2 physics server publishes atomic camera and robot feedback. Inference-time and idle supervision service heartbeats and freshness checks.
Verify and record	Contact, lift, camera placement and neighbor checks govern completion. True object state is used only for independent scoring. All images, raw proposals, traces and outcomes are saved.

Continuous execution and dependable software

Ubuntu 22.04 ARM64 / ROS2 Humble owns physics, image inference and control; the browser is only an operator display. Physics advances independently during inference and between requests. A 0.6 s active-command lease and 1.2 s communication lease latch faults when updates cease. Camera age is bounded to 1.2 wall seconds and 0.5 simulated seconds; robot feedback age is bounded to 0.8 wall seconds. Halt acknowledges the last accepted sequence and pauses simulated physics while preserving the gripper target. Reset rejects samples from the previous simulator epoch.

Physics uses 2 ms substeps with a requested 20 ms server timer. Software rendering blocks the executor: the successful development run measured 20.04 ms median tick interval, 193.53 ms p95 and 0.3835 simulation/wall-time advancement. These are not real-time or physical-arm control rates. Stop and leases are simulation behavior, not certified hardware protective functions.

The engineering progression

Question	Observed result and next decision
Can scripted contact manipulation work?	Built a baseline and 60 expert demonstrations. No object welds, attachment toggles or pose writes carry parts.
Can direct action imitation replace it?	Initial cloning 0/18; corrective cloning 0/18; recovery-augmented cloning 6/18. Failed experiments remain available.
Which learned component is dependable?	A different learned Jacobian model plus conventional feedback completed 192/240 historical tasks; 48 guarded refusals remain incomplete tasks.
Can language refer to camera pixels?	Added actual image-input Qwen3-VL, proposal guards and separate pick/hold/place. Preserved wrong model answers rather than treating all guarded refusals as model accuracy.
Can Ubuntu be responsive?	Paired fresh-image median 34.26 to 4.80 s; separate paired task 131.29 to 74.06 s with identical control trajectory and video pixels.
Can it operate under independent feedback?	Four continuous integration failures preceded a successful development repeat. Elapsed-time motion bounds, deferred preview, sampled settling and progress deadlines were validated on development scenes.

What was learned, and what was engineered

The dependable PyTorch model predicts a robot Jacobian from sine/cosine joint features using two 256-unit hidden layers. It was trained on 60000 synthetic kinematics samples with 12000 separate validation samples around training trajectories. These are not contact demonstrations. Runtime task motion uses that prediction without silently replacing it with expert IK; initial camera parking and verification holds use conventional control. This is hybrid learned-model feedback, not repaired end-to-end imitation, reinforcement learning or a VLA.

A live development session verified red pickup and follow-up placement. A longer blue hold later lost confirmed finger contact and halted; the original evidence does not prove a physical drop. Another held red task exercised explicit Stop. The first reserved protocol also exposed a motion-lease expiry during stationary inventory processing. That full protocol remains preserved. Explicit supervised sensing hold was validated on separate development scenes before a new reserved protocol; the deadline was not relaxed.

Final reserved evaluation: every case

Case / condition	Expected outcome	Observed
placement-00 / nominal	place	PASS / 137.8 s
placement-01 / nominal	place	PASS / 155.8 s
placement-02 / noise	place	PASS / 141.3 s
placement-03 / noise	place	PASS / 115.8 s
placement-04 / dim	place	PASS / 175.3 s
placement-05 / wide	place	PASS / 130.8 s
spatial-leftmost / nominal	spatial-place	PASS / 139.8 s
spatial-rightmost / nominal	spatial-place	PASS / 155.8 s
pick-followup / nominal	dialogue	PASS / 141.4 s
generic-ambiguity / nominal	clarification	PASS / 17.9 s
duplicate-red / nominal	clarification	PASS / 17.9 s
absent-red / nominal	clarification	PASS / 17.8 s
missed-grasp / gripper_fault	recovery	PASS / 173.8 s
moved-target / nominal	moved-target	PASS / 40.0 s
cancel-while-lifted / nominal	cancel	PASS / 62.6 s

Full denominator: 15/15 declared checks; 10/10 placement cases completed. The remaining five cases are three clarifications, a moved-target refusal and cancellation while lifted. They are not successful sorting placements. Seeds 99100-99114 and acceptance thresholds were saved before execution. All failures and timeouts remain in results.

Model-native abstention or independently validated selection: 12/15 attempts. Supported requests count only validated candidates; negative requests count only a raw None candidate. A guard refusal does not establish that the model itself abstained. No new localization-error bound is claimed from this evaluation.

The conditions use different object/tray pairs and are unpaired small samples. Seeds are reserved from training/development, but requests and objects are deliberately restricted. These results do not estimate reliability on arbitrary scenes, human clutter or hardware.

Verification, reproduction and ownership

Independent checks and saved evidence

Successful placement requires tray containment, support, low final velocity, released non-contacting gripper, lift above 60 mm, no monitored forbidden contact and neighbor displacement below 5 mm. The controller also verifies three camera observations. Temporary missing grasp contact permits one retry; target movement, collision, invalid commands or lost suspended grasp halt. Independent communication/command-loss probes confirmed faults latch, physics stays paused and later targets are rejected. Camera silence/replay and inference-time loss are retained in separate development evaluations.

Reproduce the portfolio

The Ubuntu setup builds pinned llama.cpp and converts the pinned Qwen checkpoint; external pretrained weights are excluded from downloads. Build/source the ROS2 interfaces, start the continuous operator with a new output directory, then run the complete frozen evaluation in an isolated ROS domain. The archive retains source history, licensed Panda assets, the trained Jacobian checkpoint, tests, original media, timelines, manifests and verification scripts. All 163 Ubuntu automated tests passed in the required OSMesa environment. Docker and hosted CI definitions are supplied but not execution-validated.

Contribution and role alignment

Google DeepMind Menagerie provides the Panda assets; Qwen supplies pretrained language/vision weights; MuJoCo, ROS2, PyTorch and llama.cpp provide their respective foundations. Workcell composition, perception, controller/supervision, training pipeline, interfaces, optimization and evaluation are project additions developed with AI-assisted coding. The project demonstrates robotics, practical vision, multimodal AI, simulation, Python, ROS2, reusable interfaces and client-style demonstration relevant to the EY Physical AI Engineer posting. It does not demonstrate physical hardware, Jetson, C++ development, RL or VLA competence.

Role reference: EY Toronto Physical AI Engineer posting. Project evidence and remaining gaps are separated in docs/role-alignment.md.

Operating scope: known opaque rigid cuboids, fixed trays and calibrated cameras. Sensors and contact are idealized. No hardware force calibration, human safety validation, arbitrary grasp planning, general-object vision or physical deployment is claimed. Continuous task phases remain engineered and this CPU simulation is slower than real time.

Evidence: runs/ubuntu-final-continuous-reserved-02; portfolio/final/film-manifest.json; docs/continuous-execution.md; docs/development-log.md; THIRD_PARTY.md.

Frozen manifest SHA256: 9c1aa6e2ef2fa913684c1a593bc1d75bc4e358398e0f159068206983a9c75ed2

Captured Python source SHA256: e0fc28c839cafe06e1b3d7dcf92014c2745370a3a85d182bd2bf394d662afd96